डॉ.श्यामाप्रसाद मुखर्जी अंतरराष्ट्रीय सूचना प्रौद्योगिकी संस्थान
DR.SHYAMA PRASAD MUKHERJEE INTERNATIONAL
INSTITUTE OF INFORMATION TECHNOLOGY, NAYA RAIPUR
(A Joint Initiative of NTPC & Govt. of Chhattisgarh)

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M.Tech - DSAI- PG curriculum

(With effect from July 2025)

Lecture-L: 1 Credit = 1 hr/ week
Tutorial-T: 1 Credit = 1hr/week
Practical-P: 1 Credit = 2 hr/week
Scholarly Engagement Hours-S: 0 Credit= 2 hr/week

M.Tech (DSAI) Programme under CM IT Fellowship Scheme: NEP 2020-aligned Course Distribution

Duration: 2 years (4 semesters) | 80 credits

Curriculum

Semester 1 Total Credit : 17 (Duration 3 Months inclusive of Weekends)					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Data Structures and Algorithm Analysis	CS-L-*xx*-C	Core	3	2–0–2-2
2	Foundations of Data Science	CS-L-*xx*-C	Core	3	2–0–2-2
3	Digital Transformation - I	CS-L-*xx*-C	Core	3	2–0–2-2
4	Program Elective I	CS-L-*xx*-E	PEC	3	3-0-0-0
5	Program Elective II	CS-L-*xx*-E	PEC	3	3-0-0-0
6	Inside Indian Governance: Policies, People, and Processes	HS-L-*xx*-C	VSEC	2	1-0-2-2
			Total	**17**	**13–0–06** **19**

Semester 2 Total Credit : 17 (Duration 3 Months inclusive of Weekends)					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Deep Learning	CS-L-*xx*-C	Core	3	2–0–2-2
2	Digital Transformation - II	CS-L-*xx*-C	Core	3	2–0–2-2
3	Program Elective III	CS-L-*xx*-E	PEC	3	3-0-0-0
4	Program Elective IV	CS-L-*xx*-E	PEC	3	3-0-0-0
5	Program Elective V	CS-L-*xx*-E	PEC	3	3-0-0-0
6	Personality Development	HS-L-*xx*-C	VSEC	2	1-0-2-2
			Total	**17**	**14–0–06** **20**

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Semester 3 Total Credit : 10 (Duration 2 Months inclusive of Weekends)					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Program Elective VI	CS-L-*xx*-E	PEC	3	3-0-0-0
2	Program Elective VII	CS-L-*xx*-E	PEC	3	3-0-0-0
3	Program Elective VIII	CS-L-*xx*-E	PEC	3	3-0-0-0
4	Industry Practices	HS-L-*xx*-C	VSEC	1	0-0-2-2
			Total	**10**	**09–0–02 11**

Semester 4 – Session 1 Total Credit : 10 (Duration 4 Months)					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Govt. Department Work Engagement followed by Seminar	CS-L-*xx*-C	EL	08	-
2	Report Writing	CS-L-*xx*-C	EL	02	
			Total	**10**	**--**

Semester 4 – Session 2 Total Credit : 13 (Duration 6 Months)					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Govt. Department Work Engagement followed by Seminar	CS-L-*xx*-C	EL	09	-
2	Report Writing	CS-L-*xx*-C	EL	04	
			Total	**13**	**--**

Semester 4 – Session 3 Total Credit : 13 (Duration 6 Months)					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Govt. Department Work Engagement followed by Seminar	CS-L-*xx*-C	EL	09	-
2	Report Writing	CS-L-*xx*-C	EL	04	
			Total	**13**	**--**

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Electives (Tentative)

Data Visualization and Analytics	Natural Language Processing	Computer Vision
Cloud Computing	Cyber Security	Financial Analytics
Government Analytics	Big Data Technologies	Human Resource Analytics
AI Assisted Programming	Remote Sensing Applications	Forensic Science and Cyber Law
Generative AI and Prompt Engineering	Human Computer Interaction	Agile Software Development

Electives are subject to change depending upon the availability of expertise, industry collaboration, and academic planning.

The institute retains the flexibility to:

- **Revise or replace elective courses** based on emerging technologies, evolving industry needs, and faculty expertise.
- Offer electives in **collaboration with industry partners**, research labs, or through **online platforms** (e.g., NPTEL, SWAYAM, Coursera) with appropriate academic integration.
- Allow students to **take interdisciplinary electives** from other departments or specializations, with prior approval and after meeting prerequisites.
- Provide opportunities for **guest lectures, micro-courses, workshops**, and **credit-bearing industry projects** in lieu of or alongside elective courses.

All changes will be communicated at the beginning of each semester through the academic office and course planning committee.